

Ticket 11 Study Report: Grid and Domain Sensitivity

SURF2026 American Option Risk Surfaces

Codex-assisted implementation report

June 16, 2026

Abstract

Ticket 11 studies whether representative American CN/PSOR outputs are stable when the grid resolution and finite spot-domain cutoff change. It uses the baseline Ticket 04 solver, not the Ticket 10A Rannacher variant. The study compares selected prices, continuation-premium boundary curves, LCP diagnostics, PSOR iterations, and finite-difference Gamma diagnostics for an American put and a dividend-paying American call. The evidence is encouraging for this preliminary ladder: obstacle violations remain zero, complementarity diagnostics remain near solver tolerance, and the proportional domain extension from $S_{\max} = 4$ to 6 does not change selected target region values. The report is intentionally cautious because three grid levels are not a formal convergence proof.

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1 Role of Ticket 11 in the Solver Project

Earlier tickets built the validation ladder: European closed-form utilities, grids and operators, European CN validation, American CN/PSOR, American put and call validation, LCP diagnostics, boundary extraction, Delta/Gamma diagnostics, and a Rannacher smoothing comparison. Ticket 10A concluded `KEEP_BASELINE_FOR_NOW`, so Ticket 11 uses the baseline CN/PSOR solver.

Ticket 11 answers a different question: do representative prices, boundaries, and diagnostics remain stable when we change M , N , and $S_{\max}$? This is required before stress maps, dataset construction, or neural surrogate modelling.

2 Theory and Methodology Connection

The documented source basis is internal: the planning report, literature map, formulation note, solver validation plan, and prior Ticket 01–10A reports. The solver validation plan explicitly places grid refinement and domain sensitivity before stress maps and neural work. This reference integration pass connects those checks to finite-difference option-PDE background (Crank and Nicolson, 1947; Wilmott et al., 1995) while preserving the ticket’s limited claim: sensitivity evidence is useful but not a proof of convergence.

3 Grid Sensitivity in Plain Language

A finite-difference solver replaces a continuous PDE problem with values on a grid. A coarse grid can miss important behavior near the strike or exercise boundary. Grid sensitivity checks whether the result changes less as the grid is refined.

In this ticket:

$$M = N \in \{80, 120, 180\}, \quad S_{\max} = 4.$$

The finest grid, $M = N = 180$, is used as the numerical reference for selected comparisons.

4 Domain Cutoff Sensitivity in Plain Language

The Black–Scholes spot domain is theoretically unbounded, but the code solves on $[0, S_{\max}]$. If $S_{\max}$ is too small, the artificial upper boundary can distort prices or boundaries inside the region we care about.

Ticket 11 tests:

$$S_{\max} \in \{4, 5, 6\}.$$

The domain cases use proportional M values:

$$(S_{\max}, M, N) \in \{(4, 120, 120), (5, 150, 120), (6, 180, 120)\}.$$

This keeps $\Delta S \approx 1/30$ comparable while changing the domain cutoff.

5 Why Ticket 11 Follows Ticket 10A

Rannacher smoothing is useful only if it is clearly reported and improves diagnostics without distorting prices or boundaries. Ticket 10A did not find a clear Gamma benefit, so Ticket 11 does not silently

switch methods. This keeps the sensitivity study tied to the already validated baseline solver.

6 Sensitivity Design and Chosen Configurations

Study	Cases	Purpose
Grid sensitivity	$M = N = 80, 120, 180,$ $S_{\max} = 4$	Check refinement behavior while holding the domain fixed.
Domain sensitivity	$S_{\max} = 4, 5, 6,$ $M = 120, 150, 180,$ $N = 120$	Check upper-boundary influence while keeping ΔS comparable.
Option families	American put and dividend-paying American call	Reuse the representative cases from Tickets 05 and 07.

7 Reference Configuration Choice

The reference is not an analytical truth. It is simply the finest or largest-domain configuration available inside Ticket 11:

- grid reference: $M = N = 180, S_{\max} = 4$;
- domain reference: $S_{\max} = 6, M = 180, N = 120$.

Every selected spot row reports actual moneyness. This matters because nearest-node comparisons can compare $S/K = 0.5$ on one grid with $S/K = 0.4888889$ on another grid.

8 Price Stability Results

Case	M, N or $S_{\max}$	Reference	Max selected diff	RMSE selected diff
american_put_grid_80	80	grid 180	0.0111111	0.00454879
american_put_grid_120	120	grid 180	0.0111111	0.00453795
american_put_grid_180	180	grid 180	0	0
dividend_call_grid_80	80	grid 180	0.0111111	0.00454255
dividend_call_grid_120	120	grid 180	0.0111111	0.00453671
dividend_call_grid_180	180	grid 180	0	0
american_put_domain_smax4	4	domain 6	0	0
dividend_call_domain_smax4	4	domain 6	0	0

The largest grid selected-spot difference is driven by nearest-node matching at $S/K = 0.5$. For example, the $M = 180$ grid's nearest spot to $0.5K$ is $0.4888889K$, while the coarser grids hit $0.5K$ exactly. This is a useful warning for beginner researchers: a selected-spot table is not the same thing as interpolation-based convergence analysis.

The domain sensitivity rows show no selected target-region price change for these configurations. Because ΔS and N are held comparable, this suggests that $S_{\max} = 4K$ is not distorting the selected target region for these representative cases.

9 Boundary Stability Results

Case	Reference	Max selected boundary shift	Interpretation
american_put_grid_80	grid 180	0.0255974	Coarse boundary visibly differs from reference.
american_put_grid_120	grid 180	0.0219485	Boundary difference remains but is moderate.
dividend_call_grid_80	grid 180	0.0220919	Coarse boundary differs from reference.
dividend_call_grid_120	grid 180	0.0217417	Moderate boundary difference.
american_put_domain_smax4	domain 6	0	Domain extension did not move selected boundaries.
dividend_call_domain_smax4	domain 6	0	Domain extension did not move selected boundaries.

Boundary curves remain financially plausible: put boundaries are below the strike, and dividend-call boundaries are above the strike. Grid-level shifts should not be overread as exact boundary convergence, because Ticket 09 extraction is threshold-based and nearest-time matching is used here.

10 LCP Diagnostic Stability Results

Case	Obstacle violation	Equation violation	Complementarity max
american_put_grid_80	0	1.30×10^{-9}	3.48×10^{-11}
american_put_grid_180	0	1.64×10^{-9}	3.25×10^{-11}
dividend_call_grid_80	0	1.67×10^{-9}	5.26×10^{-11}
dividend_call_grid_180	0	3.58×10^{-9}	2.75×10^{-11}
american_put_domain_smax6	0	1.56×10^{-9}	6.65×10^{-11}
dividend_call_domain_smax6	0	2.14×10^{-9}	2.38×10^{-11}

All PSOR steps converged across the Ticket 11 runs. LCP diagnostic magnitudes remain near solver tolerance, and no new obstacle violation appears under refinement or domain extension.

11 Greek Diagnostic Stability Results

Case	Full max $ \Gamma $	Strict max $ \Gamma $	Caution
american_put_grid_80	20	1.25684	Coarse grid damps Gamma.
american_put_grid_180	45	4.39202	Finer grid exposes sharper curvature.
dividend_call_grid_80	20	1.64950	Coarse grid damps Gamma.
dividend_call_grid_180	45	3.83192	Finer grid exposes sharper curvature.
american_put_domain_smax6	30	3.18002	Domain-stable with same ΔS .
dividend_call_domain_smax6	30	2.67989	Domain-stable with same ΔS .

The Gamma rows are diagnostic, not risk labels. The increasing full-grid maximum is expected because Γ is a second derivative and sharper grids resolve sharper kink/boundary features. This confirms Ticket 10's warning: Greek outputs require masking and careful interpretation.

12 Figure Interpretation

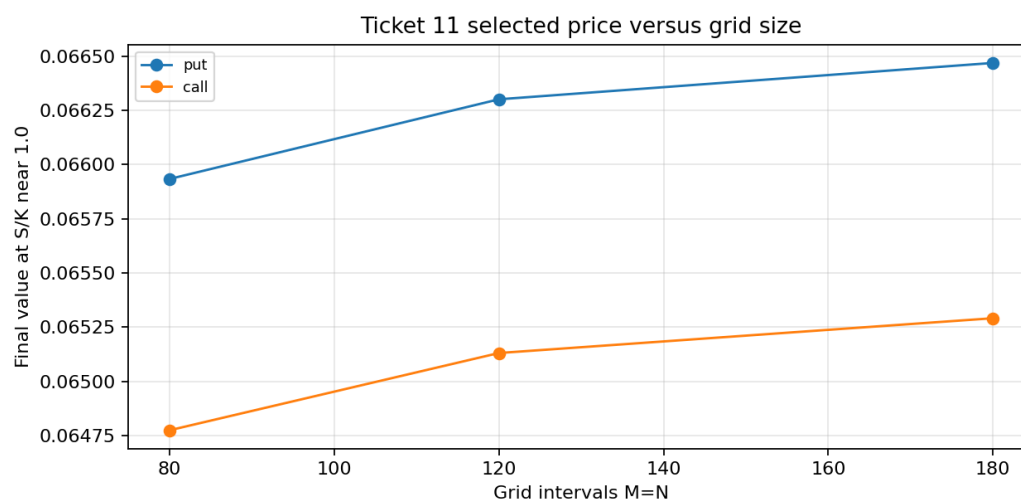


Figure 1: At-the-money selected prices versus grid size for the two representative cases.

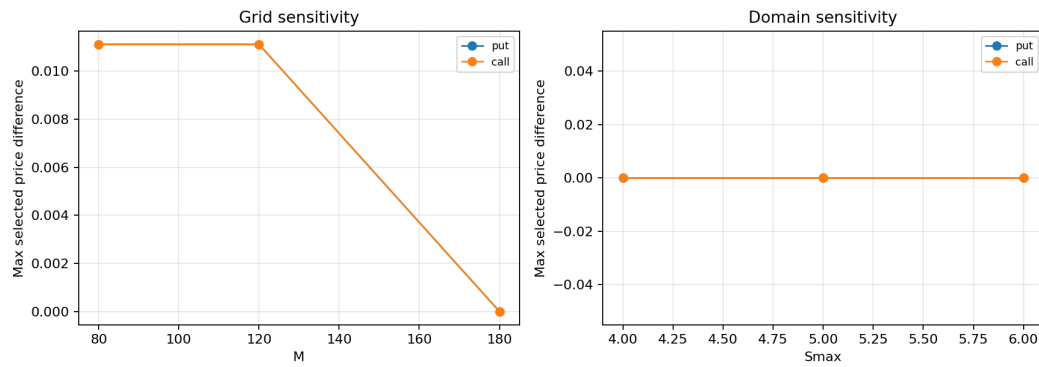


Figure 2: Selected price differences relative to the chosen grid/domain reference.

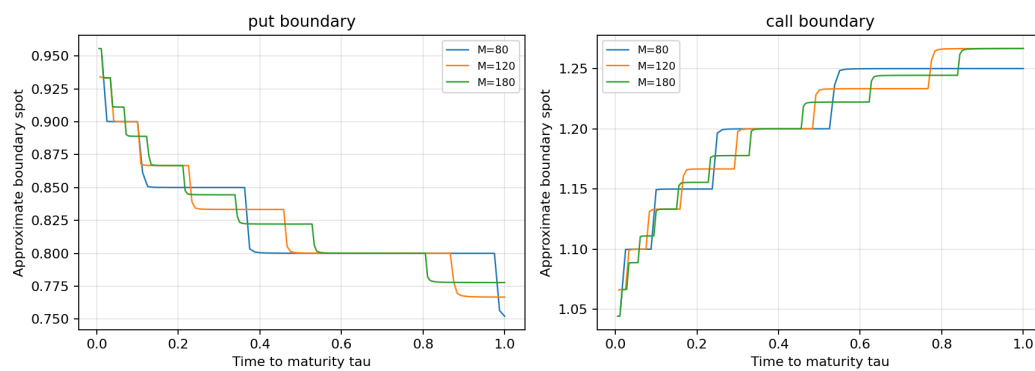


Figure 3: Approximate boundary curves across grid levels. These are threshold-based diagnostics, not exact analytical boundaries.

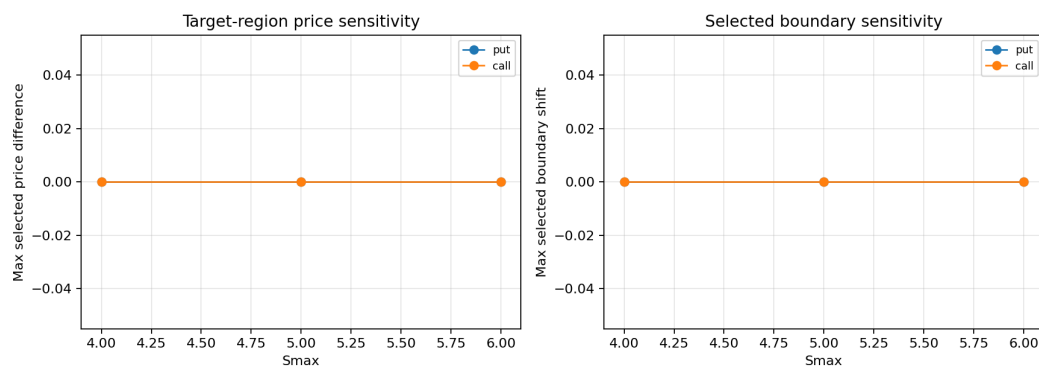


Figure 4: Domain-cutoff sensitivity is flat in the selected target region under comparable ΔS .

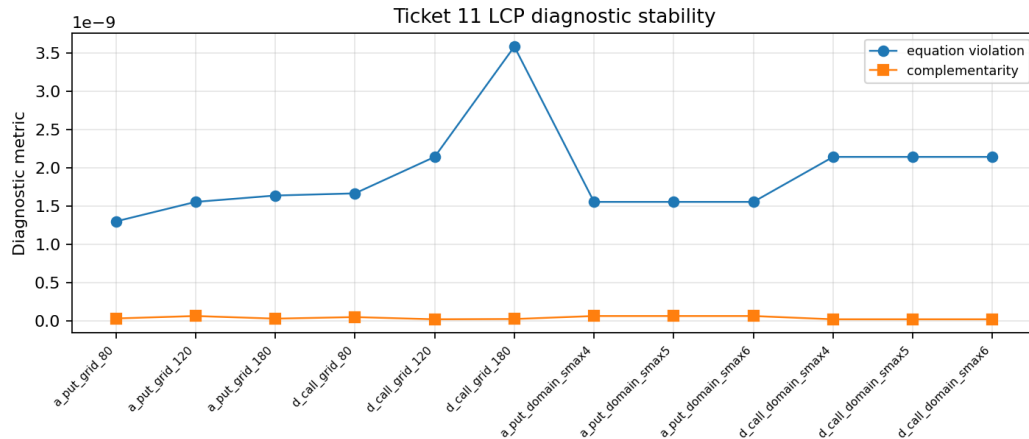


Figure 5: LCP diagnostic quantities stay near solver tolerance across sensitivity settings.

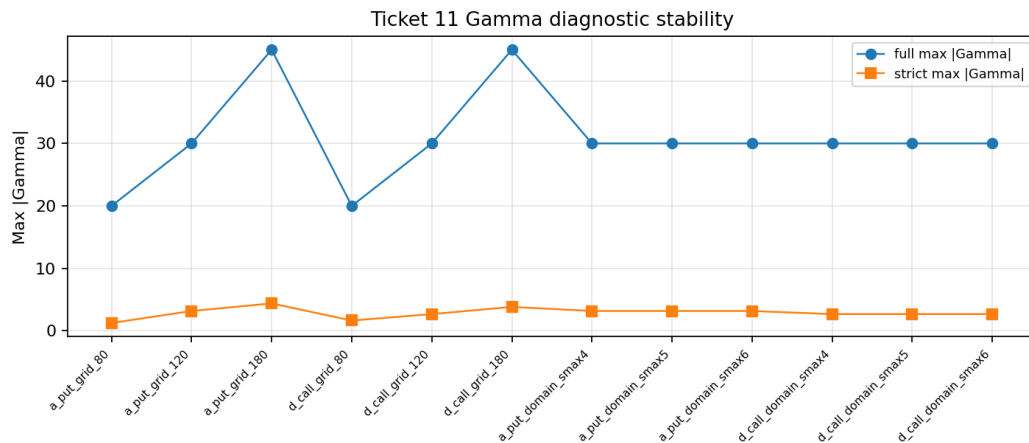


Figure 6: Gamma diagnostics are grid-sensitive, which is expected for second derivatives near non-smooth features.

13 Testing Strategy and Results

Ticket 11 tests cover configuration construction, comparable domain spacing, nearest-grid spot matching, synthetic reference differences, small real solver runs, boundary/LCP/Greek diagnostic rows, CSV artifacts, figure artifacts, scope exclusions, and Ticket 11 docstrings.

The focused Ticket 11 test command returned:

```
Ran 13 tests in 1.439s
OK
```

The full final suite is rerun after report generation and recorded in the chat completion.

14 Implementation Pitfalls and Debugging Notes

Event	Symptom	Fix	Lesson
Initial red run	Ticket 11 tests failed because <code>sensitivity.py</code> and the experiment script did not exist.	Implemented the diagnostics module and experiment.	Red tests should fail for the intended missing capability.
Intermediate run	Module tests passed, but experiment setup failed because <code>09_grid_domain_sensitivity.py</code> was still missing.	Added the experiment script and artifact writers.	Build feature layers in order: API first, then artifact orchestration.

Likely traps for future sensitivity work:

- confusing grid refinement with domain enlargement;
- comparing selected points without reporting actual moneyiness;
- overclaiming convergence from three grid levels;
- treating Gamma outputs as stable labels;
- creating stress maps before the validation ladder is reviewed.

15 Limitations

Limitation	Explanation
Only three grid levels	Enough for a preliminary sensitivity gate, not a formal convergence proof.
Nearest-node comparison	Selected prices report actual moneyiness, but interpolation is not implemented in this ticket.
Representative cases only	The study covers one put and one dividend-paying call setup.
Gamma remains fragile	Greek sensitivity is diagnostic and should not be exported as labels.
No stress maps or datasets	Ticket 11 intentionally stops before risk-surface generation.

16 Lessons Learned

For greenhand researchers, the main lesson is that numerical validation has layers. A price can look stable while a boundary or Gamma diagnostic still deserves caution. Ticket 11 shows why the project records selected spot values, actual moneyiness, boundary shifts, LCP diagnostics, and Greek metrics together.

17 Link to the Next Ticket

The next ticket can assemble the solver-validation findings into a broader review gate. Stress maps, datasets, and neural surrogate work should still wait for human review of the full validation ladder.

18 Reproducibility Record

Item	Record
Source	src/american_risk_surfaces/diagnostics/sensitivity.py.
Experiment	experiments/09_grid_domain_sensitivity.py.
Tests	tests/test_grid_domain_sensitivity.py.
Tables	Five ticket_11 CSVs under results/01_solver_validation/tables/.
Figures	Six ticket_11 PNGs under results/01_solver_validation/figures/.
Report	This L ^A T _E X source and compiled PDF.

References

J. Crank and P. Nicolson. A practical method for numerical evaluation of solutions of partial differential equations of the heat-conduction type. *Mathematical Proceedings of the Cambridge Philosophical Society*, 43(1):50–67, 1947. doi: 10.1017/S0305004100023197.

Paul Wilmott, Sam Howison, and Jeff Dewynne. *The Mathematics of Financial Derivatives*. Cambridge University Press, 1995. doi: 10.1017/CBO9780511812545.